DAY 7 — Working with NumPy: Arrays, Vectorization & Mathematical Operations

Day 7 focused entirely on **NumPy**, one of the most fundamental Python libraries for numerical computation. NumPy is the backbone of Data Science, powering operations in Pandas, Scikit-learn, TensorFlow, and many more libraries. Today's session helped me understand why NumPy is faster, more efficient, and more reliable than using Python lists.

🔹 1. Introduction to NumPy

NumPy (Numerical Python) provides:

- Multi-dimensional arrays
- Fast mathematical operations
- Broadcasting and vectorization
- Linear algebra operations
- Logical indexing
- Integration with ML frameworks

Unlike Python lists, NumPy arrays are stored in **contiguous memory**, making them extremely fast for large datasets.

🔸 2. Creating Arrays

We created arrays using:

np.array()

np.zeros()

np.ones()

np.arange()

np.linspace()

np.random.rand()

np.random.randint()

We experimented with:

- 1D arrays
- 2D matrices
- Higher-dimensional tensors

This formed the base for matrix operations required later in ML.

🔸 3. Array Indexing & Slicing

We explored:

- Index-based selection
- Slicing (arr[0:5])
- Boolean indexing (arr[arr > 10])
- Fancy indexing (arr[[1,3,5]])

These techniques make data manipulation much easier and faster.

🔸 4. Vectorized Operations

One powerful feature of NumPy is **vectorization**, which eliminates loops:

arr + 5

arr1 * arr2

np.sqrt(arr)

np.log(arr)

Vectorized operations helped me understand why ML algorithms run efficiently on NumPy arrays.

🔸 5. Mathematical & Statistical Functions

We learned:

- np.mean()
- np.median()
- np.std()
- np.sum()
- np.min() / np.max()
- np.argmin() / np.argmax()

These are essential for data preprocessing and analysis.

🔸 6. Linear Algebra Operations

NumPy contains a full linear algebra engine:

- Dot product
- Matrix multiplication
- Transpose
- Determinants
- Eigenvalues

Linear algebra is the foundation of ML algorithms, so this understanding was crucial.

Key Takeaways from Day 7

- NumPy is essential for high-speed numerical computing
- Arrays are the backbone of ML datasets
- Vectorization makes operations extremely fast
- Statistical functions help in EDA
- NumPy prepares the foundation for Pandas and ML model implementation

Day 7 gave me strong mathematical and computational confidence for upcoming ML concepts.